

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Fossil fuels & nuclear power: sustainability and new technologies - Set 1

Mark scheme

AQA spec 3.3.3, 3.3.4.1; Energy resources; textbook pp. 183-196

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Natural pressure of gas above the oil (or dissolved in it) or water beneath forces oil up the production well; pump-jacks ('nodding donkeys') increase flow (1) • Pressure falls as oil is removed so the remaining oil no longer flows to the well (1)	2
01.2	<i>1 mark for each point.</i> • Water or natural gas is pumped down an injection well to maintain the pressure and flow of oil (recovery to about 40%) (1) • Captured CO₂ can be pumped down instead, increasing oil recovery while storing the CO₂ underground (1)	2
01.3	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> • Steam injection (from combustion or solar parabolic concentrators) to heat the oil and reduce its viscosity (1) • Controlled underground combustion to heat the oil (1) • Detergents / solvents to reduce surface tension so oil flows more easily (1) • Bacteria that partially digest heavy oil into lighter oils and produce CO₂ to maintain pressure (1)	3
01.4	<i>1 mark for each point.</i> • Many wells can be drilled from a single platform, reducing land take / number of rigs (1) • Deposits under locations where rigs cannot be placed (urban areas, sensitive habitats) can be reached (1) • Drilling can follow softer strata (quicker) and target multiple small reservoirs up to 10 km away, increasing total recovery (1)	3

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187-188; AQA spec 3.3.4.1.	
02.1	<i>1 mark for each point.</i> - Shale has very low permeability so the 'tight' oil / gas cannot flow through the rock to a well (1) - Fluid is pumped down at high pressure to open fissures in the shale along which oil / gas can flow to the recovery well (1) - Sand grains (to hold fissures open) and solvents are pumped in to increase the recovery rate (1)	3
02.2	<i>1 mark for each point.</i> - Natural gas may enter aquifer water (1) - Injected chemicals may reach aquifers or the surface; toxic metals in the rock may be mobilised (1) - Large volumes of water needed (1) - Earth tremors as fracking releases natural tensions in the crust (though not tremors that could not occur naturally) (1) - Reduced by: collection, treatment and re-use of waste water; well casing standards; restricting sites away from aquifers / sensitive areas; monitoring (1)	5
02.3	<i>1 mark for each point.</i> - Extraction is expensive and energy-intensive (quarrying and hot water treatment, steam / solvent / combustion in-situ; heating shale to release oil) so it is not economically viable at current prices (1) - Reserves only include what is economically exploitable now; high environmental impacts (land take, water, CO₂) also limit exploitation (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 184, 189-190; AQA spec 3.3.4.1.	
03.1	<i>1 mark for each point.</i> - Gasification: coal (including deposits too deep to mine) is burnt underground under controlled conditions to produce fuel gases (hydrogen, carbon monoxide, methane) (1) - Allows deep coal to be exploited and gas can be piped / used in gas turbines or for hydrogen with CCS (1) - Liquefaction: coal is converted to liquid hydrocarbons directly with solvents or via gasification and chemical conversion (1) - Provides liquid fuels for vehicles / aircraft that solid coal cannot supply (1)	4
03.2	<i>1 mark for each point.</i> - A solid, ice-like crystalline structure in which methane is trapped within water ice (1) - Found at low temperatures (polar regions / permafrost) or high pressure in ocean sediments around continental margins (1)	2
03.3	<i>1 mark for each point.</i> - Water heating: hot water pumped into sediments melts the hydrate releasing methane (1) - Depressurisation: drilling reduces pressure so methane dissociates from the crystals (1) - CO₂ injection: CO₂ bonds more strongly to the ice and displaces methane (could double as CCS) (1) - Risks: methane not collected escapes to the atmosphere - powerful greenhouse gas / positive feedback; seabed ecology poorly understood; sediment instability / landslides (1)	4

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 187, 190; AQA spec 3.3.4.1.	
04.1	<i>1 mark for each point.</i> - Nuclei of large atoms (uranium-235, plutonium-239) split when hit by a neutron (1) - A small amount of mass is converted to a large amount of energy ($E = mc^2$) and more neutrons are released, causing a chain reaction (controlled by control rods / moderator) (1) - The heat boils water to steam which drives turbines and generators (1)	3
04.2	<i>1 mark for each point.</i> - Output changes slowly (days) so it cannot follow rapid fluctuations in demand (1) - Very high construction cost but low fuel cost means it is most economic running continuously at full output (1)	2
04.3	<i>1 mark for each point.</i> - Only 30 t of fuel a year (very high energy density) so no need for railways / ports to deliver millions of tonnes of coal (1) - Still need condenser cooling water (coast / large lake) and a stable site (1)	2
04.4	<i>1 mark.</i> Uranium must be mined, purified, concentrated and processed into fuel using (fossil) energy; reactor construction and decommissioning have embodied energy (1)	1
04.5	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - No commercial reactor has been fully decommissioned so costs are unknown and often exceed estimates (1) - Large projects often overrun original cost estimates (1) - Long-term waste disposal costs uncertain (1) - State subsidies may hide costs (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 191-193; AQA spec 3.3.4.1.	
05.1	<i>Any 3 from the points below - 1 mark each, maximum 3.</i> - Public opposition following accidents (Three Mile Island 1979, Chernobyl 1986, Fukushima 2011) (1) - Very complex, expensive technology needing an advanced industrial infrastructure (1) - Concerns about links between civil nuclear materials and weapons / terrorism (1) - Uncertainty over permanent disposal of radioactive waste (1) - Uncertain decommissioning costs (1)	3
05.2	<i>1 mark for each point.</i> - Only 0.7% of natural uranium is fissile U-235; the remaining 99.3% is non-fissile U-238 (a 'fertile' fuel) (1) - Neutron bombardment in the reactor converts U-238 to fissile plutonium-239 (via beta decays) (1) - So much more energy is harnessed from the mined uranium; fast reactors need no moderator but are complex and costly (1)	3

Question	Marking guidance	Marks
05.3	<i>2 marks for each point (partial credit for a partly correct point).</i> - Advantages: thorium is three times more abundant; much less radioactive waste with shorter half-lives; harder to make weapons material; no fuel enrichment needed (2) - Disadvantages: thorium-232 is not fissile so must be bred into uranium-233 (slow breeding rate); U-233 is an alpha emitter so very hazardous to handle; less developed technology with high development costs (2)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 192-194; AQA spec 3.3.4.1.	
06.1	<i>1 mark for each point.</i> - Fission: splitting the nuclei of large atoms (U-235, Pu-239) when hit by neutrons (1) - Fusion: joining the nuclei of small atoms (deuterium and tritium - isotopes of hydrogen) to form helium, releasing a neutron and energy - the energy source of stars (1)	2
06.2	<i>1 mark for each point.</i> - Deuterium (hydrogen-2) extracted from (sea)water (1) - Tritium (hydrogen-3) produced by neutron bombardment of lithium (a lithium blanket around the reactor) (1)	2
06.3	<i>1 mark for each point.</i> - Very high temperature turns hydrogen into a plasma (electrons stripped away) and gives nuclei enough kinetic energy to overcome the repulsion between positive nuclei and collide (1) - Vacuum so the plasma is not cooled by air (1) - Magnetic field holds the plasma centrally so it does not touch the container walls and cool (1)	3
06.4	<i>1 mark for each point.</i> - Fuel (deuterium / lithium) is abundant / effectively unlimited (1) - Much less radioactive waste (no long-lived fission products); no chain reaction / meltdown risk; no weapons material (1) - Not yet achieved sustained net energy output: reactors (JET) use more energy than they release; ITER aims for 500 MW out from 50 MW in; laser fusion also experimental; commercial viability decades away (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 195-196; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
07.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Fossil fuel technologies: secondary / tertiary recovery raising recovery from 20% to 60%; directional drilling and subsea wells; ROVs; fracking for tight oil and gas; tar sands / oil shales; coal gasification / liquefaction; enhanced gas recovery; methane hydrates; CCS (pre / post combustion, storage) • Limits: still finite (discoveries declining since the 1980s); rising energy and financial cost of unconventional sources; environmental impacts (aquifers, water use, tremors, land take, habitat damage); CO₂ and other pollution; CCS unproven at scale and impractical for vehicles • Nuclear: new uranium extraction (polymer adsorption from seawater, phosphate mining, coal ash); molten salt reactors; plutonium breeder and thorium reactors increasing fuel use many-fold; fusion as a long-term prospect • Limits: cost overruns, decommissioning, waste, public opposition, proliferation, embodied energy of low-grade uranium; fusion not commercial • Sustainability criteria: maintaining supply without unacceptable environmental, economic or social impact; comparison with renewables and energy conservation; energy gap argument • Judgement: technologies can extend supplies and reduce some impacts but cannot make non-renewables sustainable indefinitely; role as transition / base-load alongside renewables	9
07.2	<i>1 mark.</i> • 80-90% (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 183-196; AQA spec 3.3.3-3.3.4.1.	